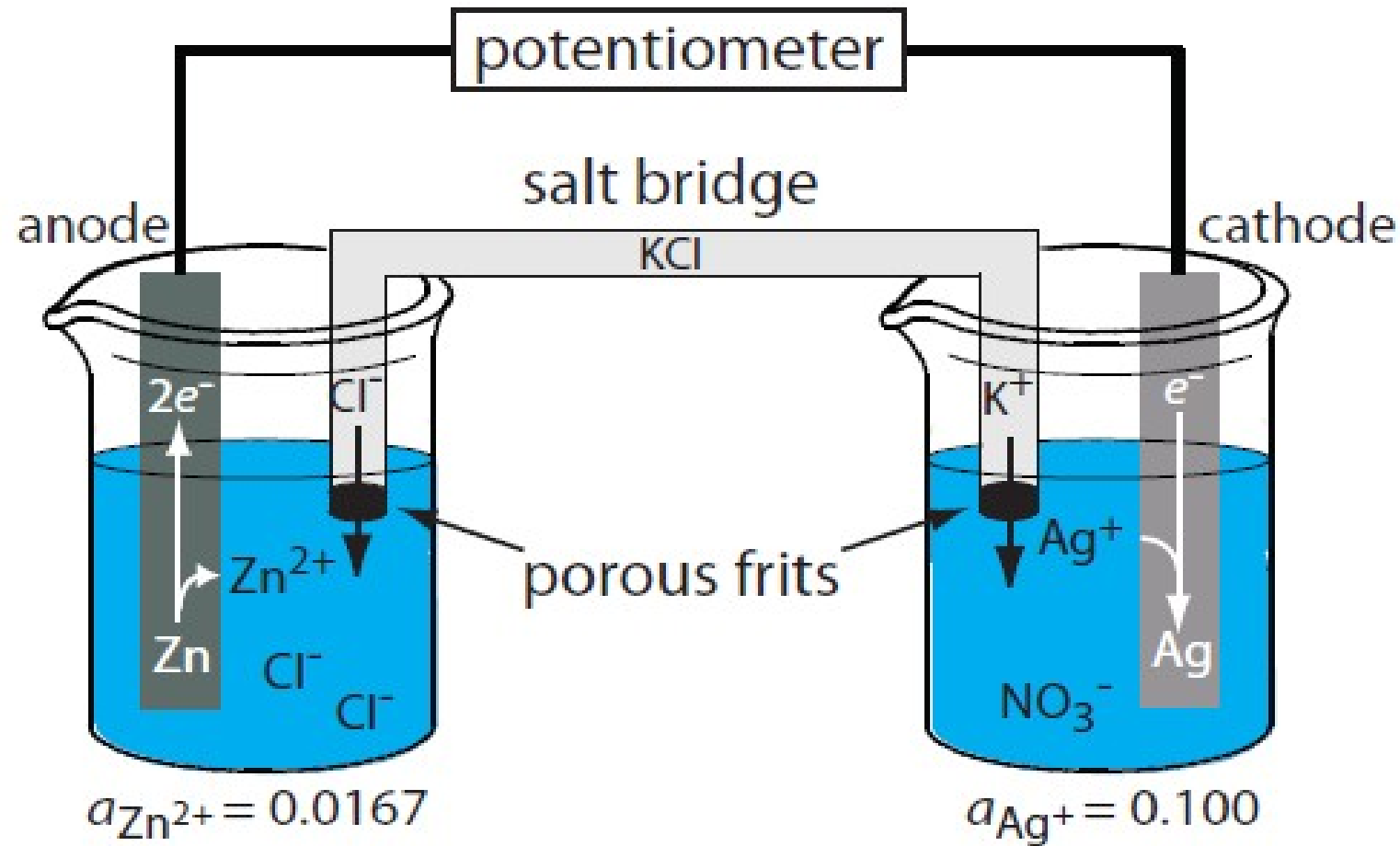


Potentiometry

Potentiometric Methods

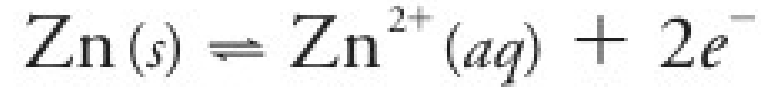
- **In potentiometry we measure the potential of an electrochemical cell under static conditions.**
- **Because no current—or only a negligible current—flows through the electrochemical cell, its composition remains unchanged.**
- **For this reason, potentiometry is a useful quantitative method of analysis.**



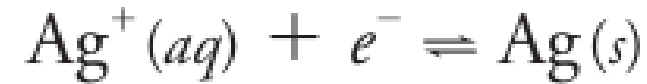
- We use a potentiometer to determine the difference between the potential of two electrodes.

- The potential of one electrode—**the working or indicator electrode**—**responds to the analyte's activity** .
- Other electrode—**reference electrode**—**has a known, fixed potential**.
- The electrochemical cell consists of two half-cells, each of which contains an electrode immersed in a solution of ions whose activities determine the electrode's potential.
- A salt bridge that contains an inert electrolyte, such as KCl, connects the two half-cells.

- **Recall,**
- **Electrode on the left as the anode and assign to it the oxidation reaction;**



- **The electrode on the right is the cathode, where the reduction reaction occurs**



- **Potential of the electrochemical cell in above figure is for the reaction**



- **NB. The cathode is the indicator electrode and the anode is the reference electrode.**

Potential and activity—the Nernst equation

- **The potential of a potentiometric electrochemical cell is,**

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

- where E_{cathode} and E_{anode} are reduction potentials for the redox reactions at the cathode and the anode, respectively.
- **Each reduction potential are is by the Nernst equation**

$$E = E^{\circ} - \frac{RT}{nF} \log Q$$

$$E = E^{\circ} - \frac{0.05916}{n} \log Q \dots \text{at } 25^{\circ}\text{C}$$



- Q is given by

$$Q \approx \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

- **The potential of the anode and cathode,**

$$E_{\text{anod}} = E^{\circ}_{\text{Zn}^{2+}/\text{Zn}} - \frac{0.05916}{2} \log \frac{1}{a_{\text{Zn}^{2+}}}$$

$$E_{\text{cathod}} = E^{\circ}_{\text{Ag}^{+}/\text{Ag}} - \frac{0.05916}{1} \log \frac{1}{a_{\text{Ag}^{+}}}$$

- Substituting E_{cathode} and E_{anode} into equation, along with the activities of Zn^{2+} and Ag^{+} and the standard-state reduction potentials, gives E_{cell} as

$$E_{\text{cell}} = \left[E^{\circ}_{\text{Ag}^{+}/\text{Ag}} - \frac{0.05916}{1} \log \frac{1}{a_{\text{Ag}^{+}}} \right] - \left[E^{\circ}_{\text{Zn}^{2+}/\text{Zn}} - \frac{0.05916}{2} \log \frac{1}{a_{\text{Zn}^{2+}}} \right]$$

NB: If there is a pure liquid, **pure solid, or the solvent**, it is not included in the equation due to the fact that its concentration is constant

$$E_{\text{cell}} = \left[0.7996\text{V} - \frac{0.05916}{1} \log \frac{1}{0.100} \right] - \left[0.7618\text{V} - \frac{0.05916}{2} \log \frac{1}{0.0167} \right]$$

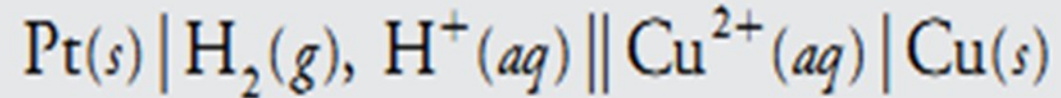
$$= +1.555\text{V}$$

Exercise 1

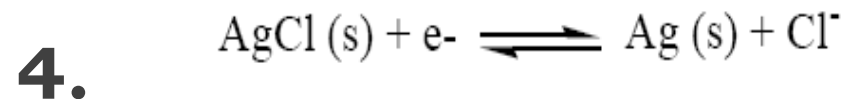
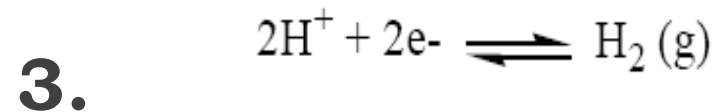
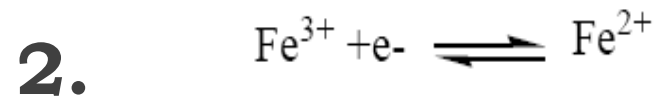
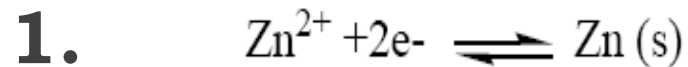
What is the activity of Fe^{3+} in an electrochemical cell similar to that in above example if the activity of Cl^- in the left-hand cell is 1.0, the activity of Fe^{2+} in the right-hand cell is 0.015, and E_{cell} is +0.546 V?

Exercise:2

A. Write the reactions occurring at the anode and the cathode for the potentiometric electrochemical cell with the following shorthand notation.

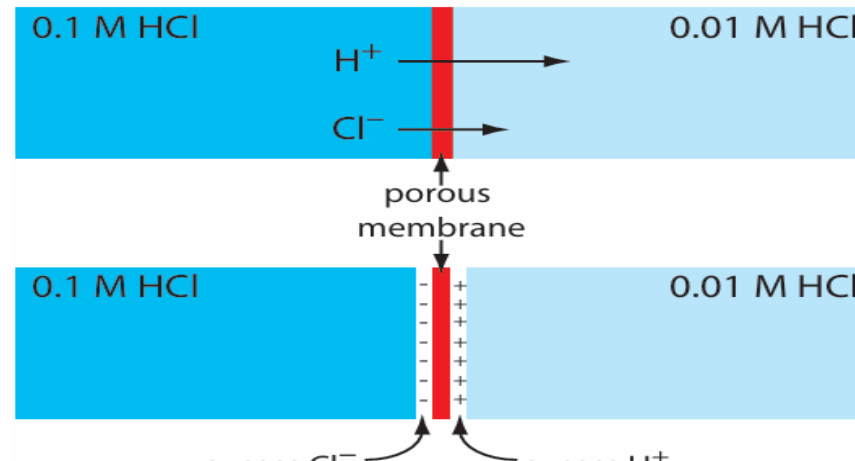


B. write the nernst equation for the ff.



Junction Potentials

- A junction potential develops at the interface between two ionic solutions if there is a difference in the concentration and mobility of the ions.
- Consider, for example, a porous membrane that separates a solution of 0.1 M HCl from a solution of 0.01 M HCl.
- Because the concentration of HCl on the membrane's left side is greater than that on the right side of the membrane, H^+ and Cl^- will diffuse in the direction of the arrows.



- Mobility of H^+ , however, is greater than that for Cl^- , as shown by the difference in the lengths of their respective arrows.
- Because of this difference in mobility, the solution on the right side of the membrane develops an excess concentration of H^+ and a positive charge.
- Simultaneously, the solution on the membrane's left side develops a negative charge because there is an excess concentration of Cl^- .
- We call this difference in potential across the membrane a junction potential and represent it as E_j

- The magnitude of the junction potential depends upon the difference in the concentration of ions on the two sides of the interface.
- When we measure the potential of an electrochemical cell, the junction potential also contributes to E_{cell} ; thus, we rewrite.

$$E_{\text{cell}} = E_{\text{c}} - E_{\text{a}} + E_{\text{j}}$$

- We also can minimize the junction potential by incorporating a high concentration of the salt in the salt bridge.
 - for which the mobilities of the cation and anion are approximately equal.

Types of electrodes

1. Reference electrode

2.Indicator electrode

- **Metallic indicator**
- **Ion selective (membrane)electrode**
- **Glass electrode**

1. Reference Electrodes

- In an electrochemical cell one of the two **half-cells provides a fixed reference potential** and the potential of the other **half-cell responds to the analyte's concentration**.
- Potentiometric electrochemical cells are constructed such that one of the half-cells provides a **known reference potential (reference electrode)**.
- The potential of the other half-cell **indicates the analyte's concentration (indicator electrode)**.

- By convention, the reference electrode is the anode; thus, the short hand notation for a potentiometric electrochemical cell is,

Reference electrode || Indicator electrode

and the cell potential is

$$\mathbf{E_{cell} = E_{ind} - E_{Ref} + E_j}$$

Properties of an ideal RE

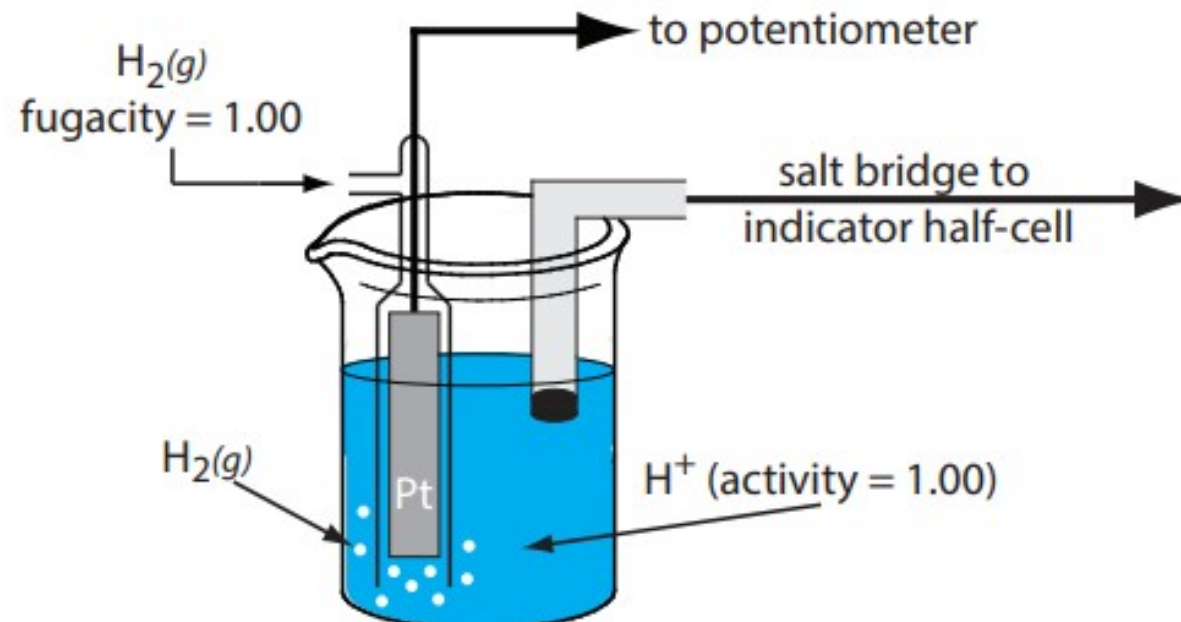
- It **must provide a stable potential** so that any change in E_{cell}
 - *is attributed to the indicator electrode, and, therefore, to a change in the analyte's concentration.*
- Ideally, **if a small current is passed through the electrode, the potential change is negligible.**
 - and in any case, returns to the initial value when the current ceases.
- In addition, the ideal reference electrode **should be easy to make and to use.**
 - Should be reproducible from electrode to electrode.

cont..

- The **ideal reference electrode provides a stable, known potential** so that we can attribute any change in E_{cell} is due to the analyte's effect on the [indicator electrode's potential](#).
- **Three common reference electrodes are discussed below.**

A. Standard hydrogen electrode(SHE)

- It is the reference electrode used to **establish standard-state potentials** for other half-reactions.
- The SHE consists of a **Pt electrode immersed in a solution** in which the activity of hydrogen ion is 1.00 and in which the fugacity of $\text{H}_2(\text{g})$ is 1.00. Look the figure below.



Cont..

- A conventional **salt bridge** connects the SHE to the **indicator half-cell**. The **short hand notation** for the standard hydrogen electrode is,



and **the standard-state potential for the reaction,** $\text{H}^+(\text{aq}) + e^- \rightleftharpoons \frac{1}{2}\text{H}_2(\text{g})$ is,

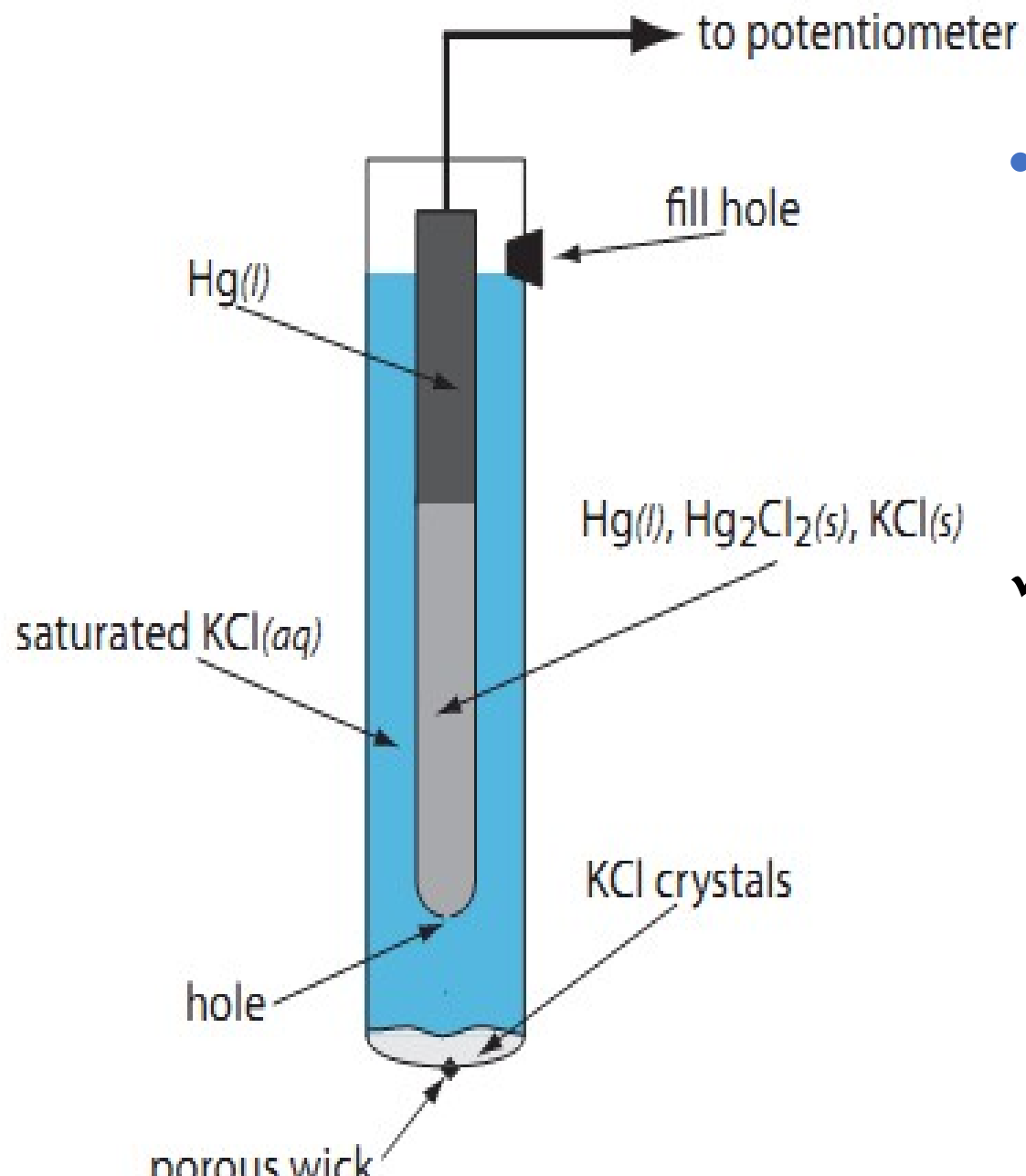
by definition, 0.00 V at all temperatures.

Cont..

- Despite its importance as the fundamental reference electrode against which we measure all other potentials,
 - *The SHE is rarely used because it is difficult to prepare and inconvenient to use.*

B. Saturated calomel electrode (SCE):

- Calomel is the common name for the compound **Hg₂Cl₂**.
- SCE consists of two concentric glass tubes
 - inner tube packed with a paste of Hg, Hg₂Cl₂, and KCl
 - a second tube that contains a saturated solution of KCl.
- **A small hole connects the two tubes and a porous wick serves as a salt bridge to the solution in which the SCE is immersed.**



- The short hand notation for this cell is,

- ✓ Because the concentration of Cl^- is fixed by the solubility of KCl , the potential of an SCE remains constant.

- The potential of the SCE is $+0.2444 \text{ V}$ at 25°C .

- A calomel reference electrode is based on the following redox couple between Hg_2Cl_2 and Hg



for which the potential is

$$E = E^\circ_{\text{Hg}_2\text{Cl}_2/\text{Hg}} - \frac{0.05916}{2} \log(a_{\text{Cl}^-})^2$$

$$= +0.2682\text{V} - \frac{0.05916}{2} \log(a_{\text{Cl}^-})^2$$

- The potential of a calomel electrode, therefore, depends on the activity of Cl^- in equilibrium.
 - However, concentration of Cl^- is fixed and the potential of the SCE remain constant (i.e. +0.2444 V at 25°C).

C. Silver/silver chloride electrodes

- Another common reference electrode is the **silver/silver chloride electrode**, which is based on the reduction of AgCl to Ag.

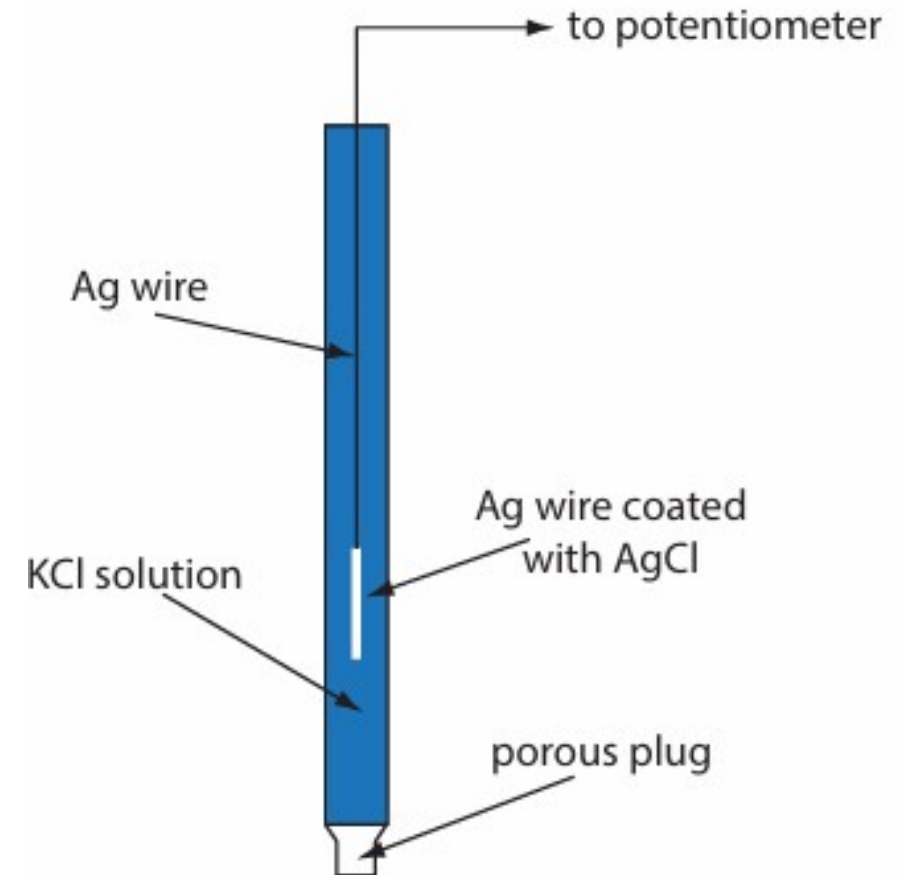


- As is the case for the calomel electrode, the activity of Cl^- determines the potential of the Ag/AgCl electrode; thus.

$$\begin{aligned} \bullet \quad E &= E^\circ_{\text{AgCl/Ag}} - 0.05916 \log a_{\text{Cl}^-} \\ &= +0.2223 \text{ V} - 0.05916 \log a_{\text{Cl}^-} \end{aligned}$$

- When prepared using a saturated solution of KCl, the electrode's potential is +0.197 V at 25 °C.

- A typical Ag/AgCl electrode consists of a **silver wire**, the end of which is coated with a **thin film of AgCl**, **immersed in a solution that contains the desired concentration of KCl**.
- A porous plug serves as the salt bridge.
- The electrode's short hand notation is,



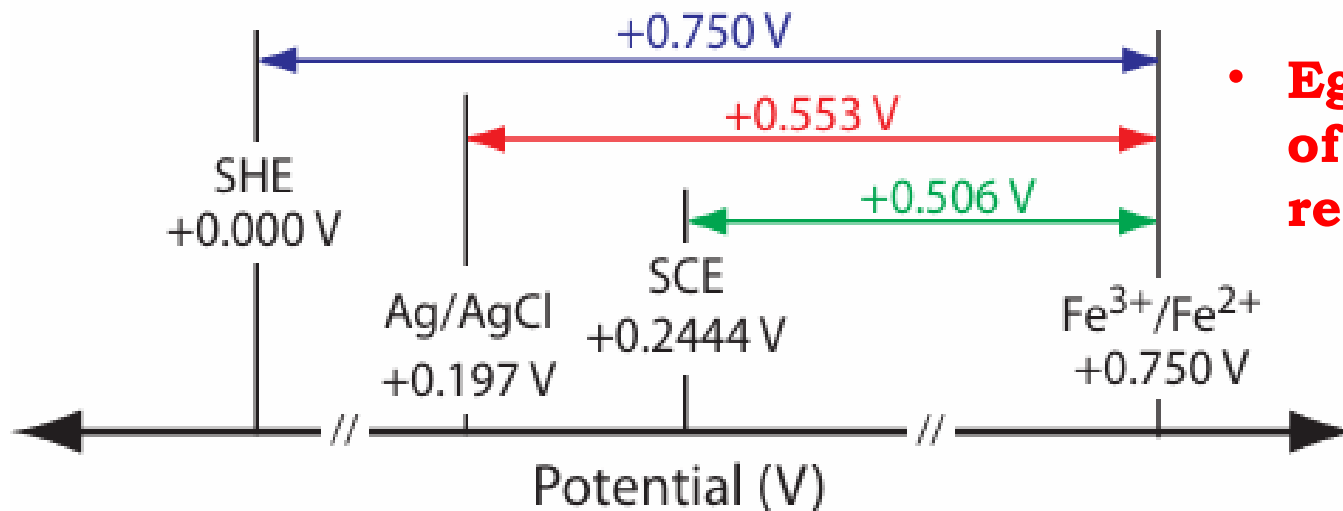
Converting Potentials Between reference electrodes

- The standard state reduction potentials in most tables are reported relative to the standard hydrogen electrode's potential of +0.00 V.
- **Reduction potential:** the magnitude of the half cell's tendency to acquire electrons and **proceed as a reduction.**
- If **done under standard conditions**, then it is referred to as the standard reduction potential.

Table 1: Standard reduction potentials

			E_o
$\text{Ce}^{4+} + \text{e}$	$\longrightarrow$	Ce^{3+}	1.61 V
$\text{MnO}_4^- + 5\text{e} + 8\text{H}^+$	$\longrightarrow$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\text{Cl}_2 + 2 \text{e}$	$\longrightarrow$	2Cl^-	1.36 V
$\text{Br}_2 + 2 \text{e}$	$\longrightarrow$	2Br^-	1.065 V
$\text{Fe}^{3+} + \text{e}$	$\longrightarrow$	Fe^{2+}	0.771 V
$\text{I}_2 + 2 \text{e}$	$\longrightarrow$	2I^-	0.536 V
$\text{AgCl} + \text{e}$	$\longrightarrow$	$\text{Ag} + \text{Cl}^-$	0.223 V
$2\text{H}^+ + 2 \text{e}$	$\longrightarrow$	H_2	0 V
$\text{Fe}^{2+} + 2 \text{e}$	$\longrightarrow$	Fe	-0.440 V
$\text{Ca}^{2+} + 2 \text{e}$	$\longrightarrow$	Ca	-2.888 V

- Because we **rarely use the SHE as a reference electrode**, we need to convert an indicator electrode's potential to its equivalent value when using a different reference electrode.



- Eg. Relationship between the potential of an $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell relative to the reference electrodes.**

- The potential relative to a standard hydrogen electrode is shown in **blue**, the potential relative to a saturated silver/silver chloride electrode is shown in **red**, and the potential relative to a saturated calomel electrode is shown in **green**.

Example

- The potential for an $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell is +0.750 V relative to the standard hydrogen electrode. What is its potential if we use a saturated calomel electrode or a saturated silver/silver chloride electrode?
- **Solution**: When we use a SHE the potential of the electrochemical cell is

$$E_{\text{cell}} = E_{\text{Fe}^{3+}/\text{Fe}^{2+}} - E_{\text{SHE}} = 0.750\text{V} - 0.000\text{V} = +0.750\text{V}$$

- We can use the same equation for a saturated calomel electrode:

$$E_{\text{cell}} = E_{\text{Fe}^{3+}/\text{Fe}^{2+}} - E_{\text{SCE}} = 0.750\text{V} - 0.2444\text{V} = +0.506\text{V}$$

or a saturated silver/silver chloride electrode

$$E_{\text{cell}} = E_{\text{Fe}^{3+}/\text{Fe}^{2+}} - E_{\text{AgCl/Ag}} = 0.750\text{V} - 0.197\text{V} = +0.553\text{V}$$

Exercise

- The potential of a $\text{UO}_2^+ / \text{U}^{4+}$ half-cell is -0.0190 V relative to a saturated calomel electrode. What is its potential when using a saturated silver/ silver chloride electrode or a standard hydrogen electrode?

2. Indicator electrodes

- Measure the **potential of the analyte solution** comparing with that of the **ref. electrode**
- Their potential are directly proportional to the ion conc.
- **Two general types:**
 - Metallic electrodes
 - Membrane (ion-selective) electrodes

A.Metal Indicator Electrodes

- **Develop an electric potential as a consequence of a **redox process** occurring at the metal surface.**
- **Three types of metallic electrodes**
 - i. Electrodes of the First Kind*
 - ii. Electrodes of the second kind*
 - iii. Electrodes of the third kind*

1. Electrodes of the First Kind

- An electrode of the first kind is a **pure metal electrode** that is in direct equilibrium with its **cation in solution**.
- For example, the equilibrium between a metal M and its cation M^{n+} is



- **Composed of the metal rod immersed in its metal solution.**
- **If we place a copper electrode in a solution that contains Cu^{2+} , the electrode's potential due to the reaction,**



is determined by the **activity of Cu^{2+}**

$$E = E^{\circ}_{\text{Cu}^{2+}/\text{Cu}} - \frac{0.05916}{2} \log \frac{1}{a_{\text{Cu}^{2+}}}$$

$$= +0.3419\text{V} - \frac{0.05916}{2} \log \frac{1}{a_{\text{Cu}^{2+}}}$$

- If **copper is the indicator electrode** in a potentiometric electrochemical cell that also includes **a saturated calomel reference electrode**.



- Then we can use the cell potential to determine an unknown activity of Cu^{2+} in the **indicator electrode's half-cell**.

$$E_{\text{cell}} = E_{\text{ind}} - E_{\text{SCE}} + E_j$$

$$= +0.3419\text{V} - \frac{0.05916}{2} \log \frac{1}{a_{\text{Cu}^{2+}}} - 0.2224\text{V} + E_j$$

- In general, if a metal, M, is in a solution of M^{n+} , the cell potential is.

$$E_{\text{cell}} = K - \frac{0.05916}{n} \log \frac{1}{a_{M^{n+}}}$$

$$= K - \frac{0.05916}{n} \log \frac{a_{M^{n+}}}{1}$$

- Where, K is a constant that includes the standard-state potential for the M^{n+}/M redox couple, the potential of the reference electrode, and the junction potential.
- ✓ Note that including E_j in the constant K means we do not need to know the junction potential's actual value; however the junction potential must remain constant if K is to maintain a constant value.

ii. Electrodes of the second kind

- When an electrode of ***the first kind responds to the potential of another ion***
➡ that is in equilibrium with M^{n+} , it is called an **electrode of the second kind**.
- E.g: Silver works as **halide indicator** electrode if coated with **silver halide (in this case AgI)**.
- If we incorporate this electrode into a potentiometric electrochemical cell with a saturated calomel electrode.



then the cell potential is

$$E_{\text{cell}} = K - 0.05916 \log a\text{I}^{-}$$

iii. Electrodes of the third Kind (redox electrodes)

- It is inert metal electrode immersed in the redox solution.
- Inert metals transport electrons to & from solution for a redox half reaction.



- The electrode itself does not undergo oxidation or reduction.
- While 1st/2nd undergo oxidation or reduction

B. Membrane Electrodes

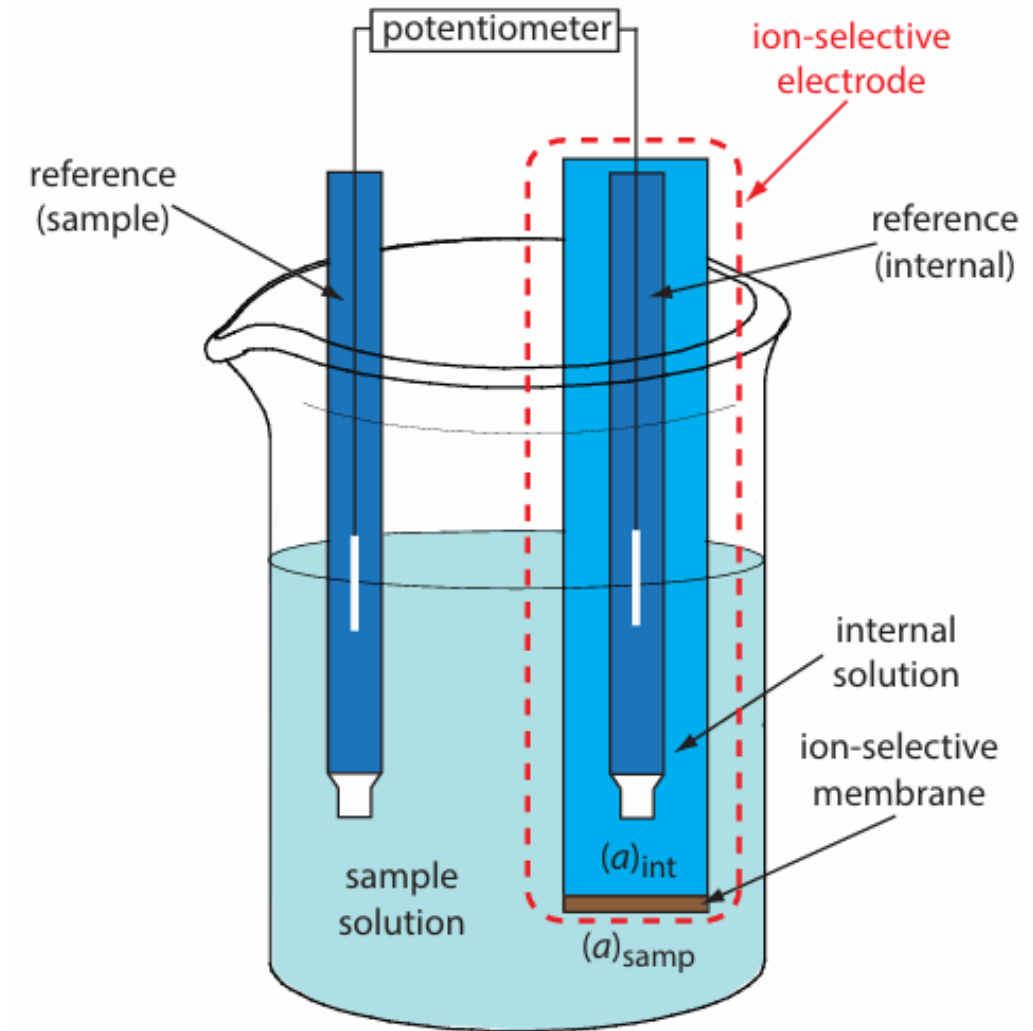
- **Ion selective electrode (ISE)**: An electrode in which the **membrane potential** is a function of the **concentration** of a particular **ion in solution**.
- **Membrane potential**
 - A potential developing across a **conductive membrane** whose **opposite sides** are in contact with solutions of **different composition**.

- Diagram for a potentiometric electrochemical cell equipped with an **ion-selective electrode**.

- The shorthand notation for this cell is

Ref(samp) || [A]samp | [A]int | | Ref(int)

Where.: the membrane is represented by the vertical slash (|) separating the two solutions containing analyte.



- **Two reference electrodes** are used; one positioned within the **internal solution**, and **one in the sample solution**.

- **The cell potential, therefore, is**

$$E_{\text{cell}} = E_{\text{Ref(int)}} - E_{\text{Ref(samp)}} + E_{\text{mem}} + E_{\text{lj}}$$

✓ where E_{mem} is the potential across the membrane.

- Since the **liquid junction potential** and **reference electrode potentials** are **constant**, any change in the cell's potential is attributed to the **membrane potential**.

- **One side of the membrane** is in contact with an **internal solution** containing a **fixed concentration of analyte**, while the other side of the **membrane is in contact with the sample**.
- Interaction of the analyte with the membrane results in a **membrane potential** if there is a difference in the analyte's concentration on **opposite sides of the membrane**.
- The membrane potential is given,

$$E_{\text{cell}} = K + \frac{0.05916}{z} \log[A]_{\text{samp}}$$

- ✓ where $[A]_{\text{samp}}$ and $[A]_{\text{int}}$ are the concentrations of analyte in the sample and the internal solution, respectively, and z is the analyte's charge.

- Ideally, E_{mem} should be zero when the concentrations of analyte on both sides of the membrane are equal.
- The term E_{asym} , which is called an asymmetry potential, accounts for the fact that the membrane potential is usually not zero under these conditions.
- Substituting the above and equation rearranging gives the,

$$E_{cell} = K + \frac{0.05916}{z} \log[A]_{samp}$$

- General equation, and applies to all types of ion-selective electrodes.

❖ Where K is a constant for the potentials of the reference electrodes, any liquid junction potentials, the asymmetry potential, and the concentration of analyte in the internal solution.

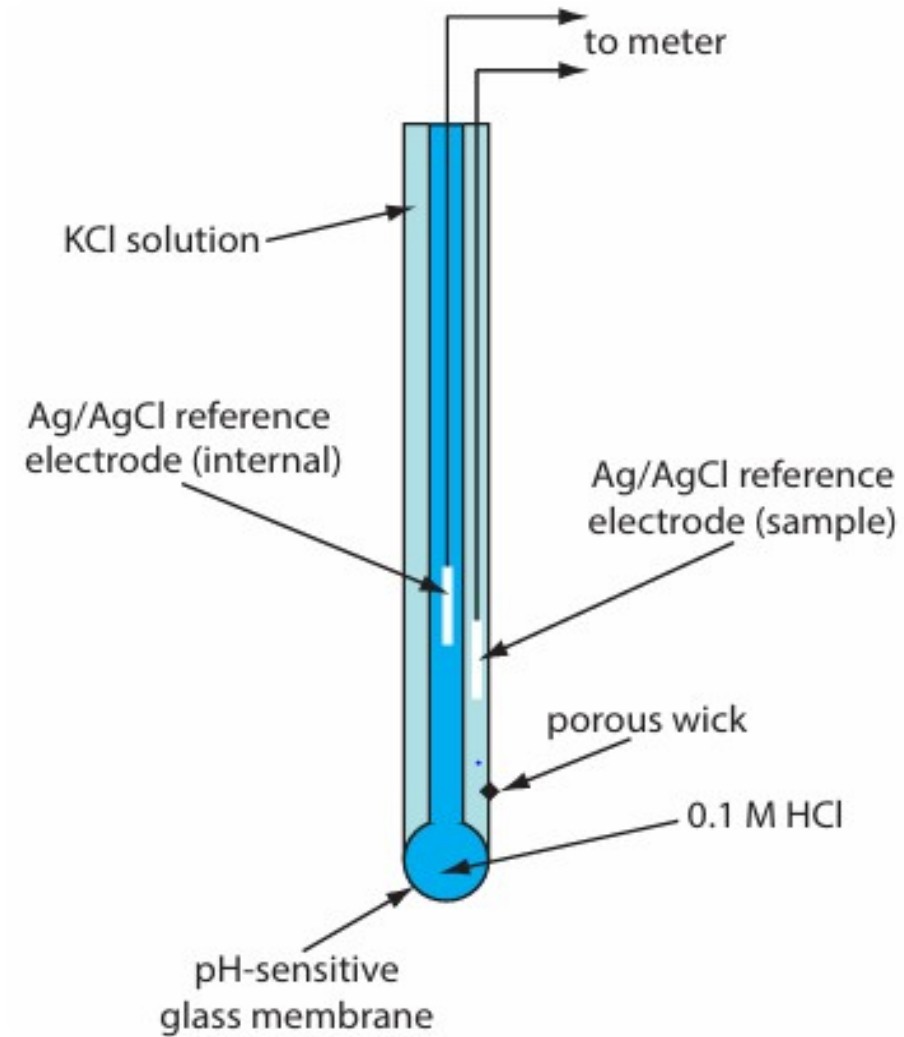
C. Glass Electrodes

- **The glass electrode is the most widely used hydrogen ion responsive electrode**
- **Its use is dependent upon the fact that when a glass membrane is immersed in a solution.**
 - *a potential is developed which is a linear function of hydrogen ion concentration of the solution.*
- **The basic arrangement of a glass electrode is shown in the figure below**

- o The bulb is immersed in the solution whose **hydrogen ion concentration** is to be measured.
- o The electrical circuit is completed by filling the bulb with a solution of **hydrochloric acid** and inserting a silver-silver chloride electrode.
- The overall potential of the electrode is governed by the **hydrogen ion concentration of the test solution**.

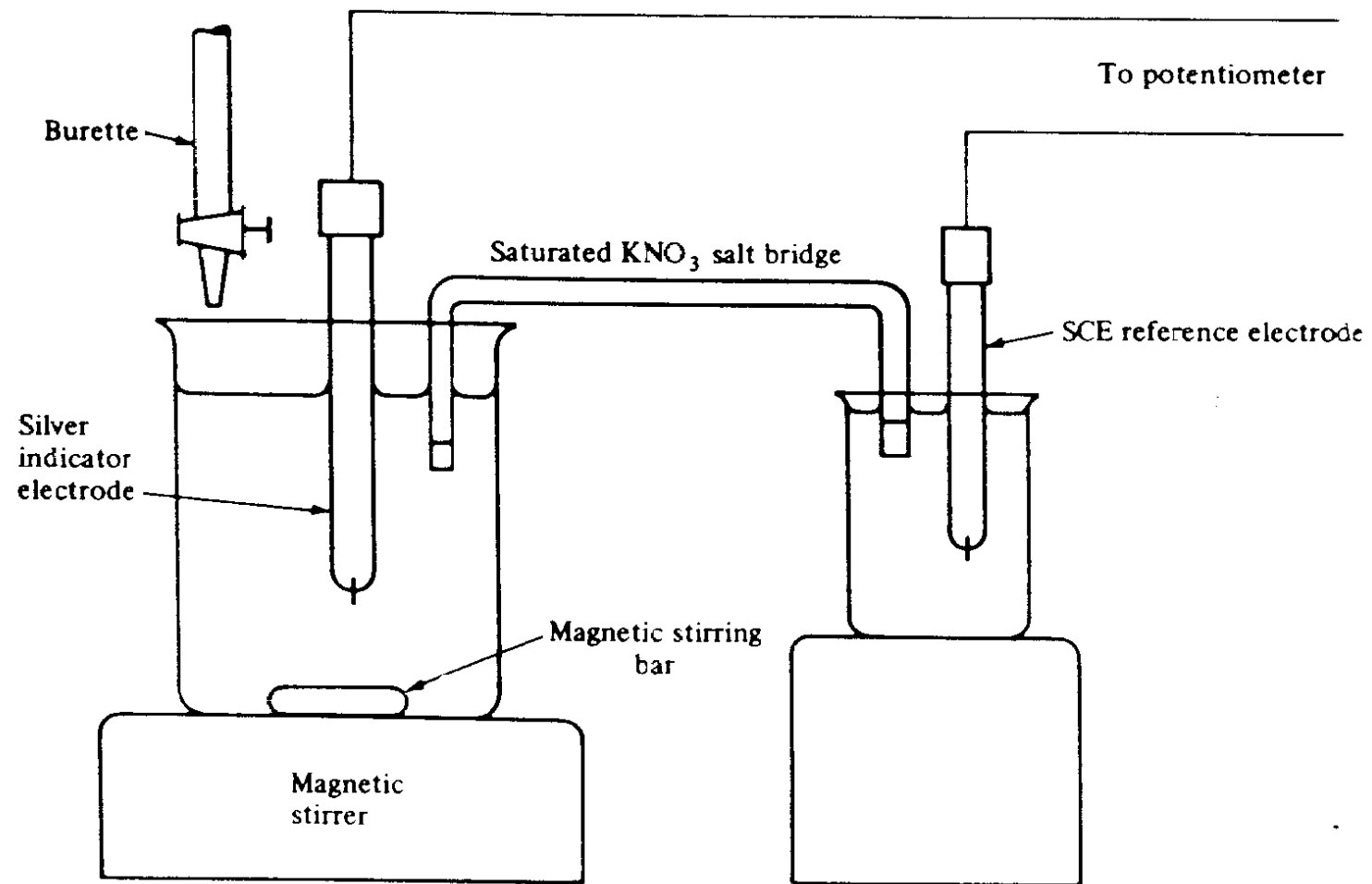
o **The potential of glass electrodes:**

$$E_{\text{cell}} = K + 0.05916 \log [\text{H}^+]$$



Application of potentiometry

- Direct potentiometry
 - Determination of sample concentration using Nernst equation
- Measuring pH of solution
- Potentiometric titration
 - Determine **end point of titration**



Typical arrangement for the potentiometric titration of chloride ion by silver(I) ion.

Application cont...

- **It is used for all types of volumetric analysis: acid base, precipitometry, complexometry and redox**
- **It is used when it is not easy or impossible to detect the end point by ordinary visual methods i.e:**
 1. **For highly coloured or turbid solutions**
 2. **For very dilute solutions**
 3. **When there is no available indicator**

a) Neutralization reactions:

- ✓ **Indicator electrode:** glass electrode
- ✓ **Reference electrode:** calomel electrode for determination of pH
- ✓ **The potential of such system can be measured by**

$$E = k - 0.0592 \text{ pH}$$

where k is the asymmetry potential, a constant which depends on the electrode system used.

b).Precipitation reactions:

- ✓ **Indicator electrode: Silver wire electrode**
- ✓ **Reference electrode: saturated calomel / Ag
Agcl/hydrogen electrode**

$$E = E^{\circ} + \frac{0.0592}{n} \log [M^{n+}]$$

where [M] - refers to the concentration of ions.

n - electronic state

a) Complex formation titration:

- ✓ metal and membrane electrodes for determination of many cations (using EDTA)
- ✓ **Indicator electrode:** silver / mercury electrode
- ✓ **Reference electrode:** saturated calomel electrode

b) Redox titration:

- ✓ **Indicator electrode:** Platinum wire or foil
- ✓ **Reference electrode:** saturated calomel / Ag-Agc electrode
- ✓ **The potential of such system can be measured by**

$$E = E^{\circ} + \frac{0.0592}{n} \log \frac{[\text{ox}]}{[\text{Red}]}$$

where E° - standard potential

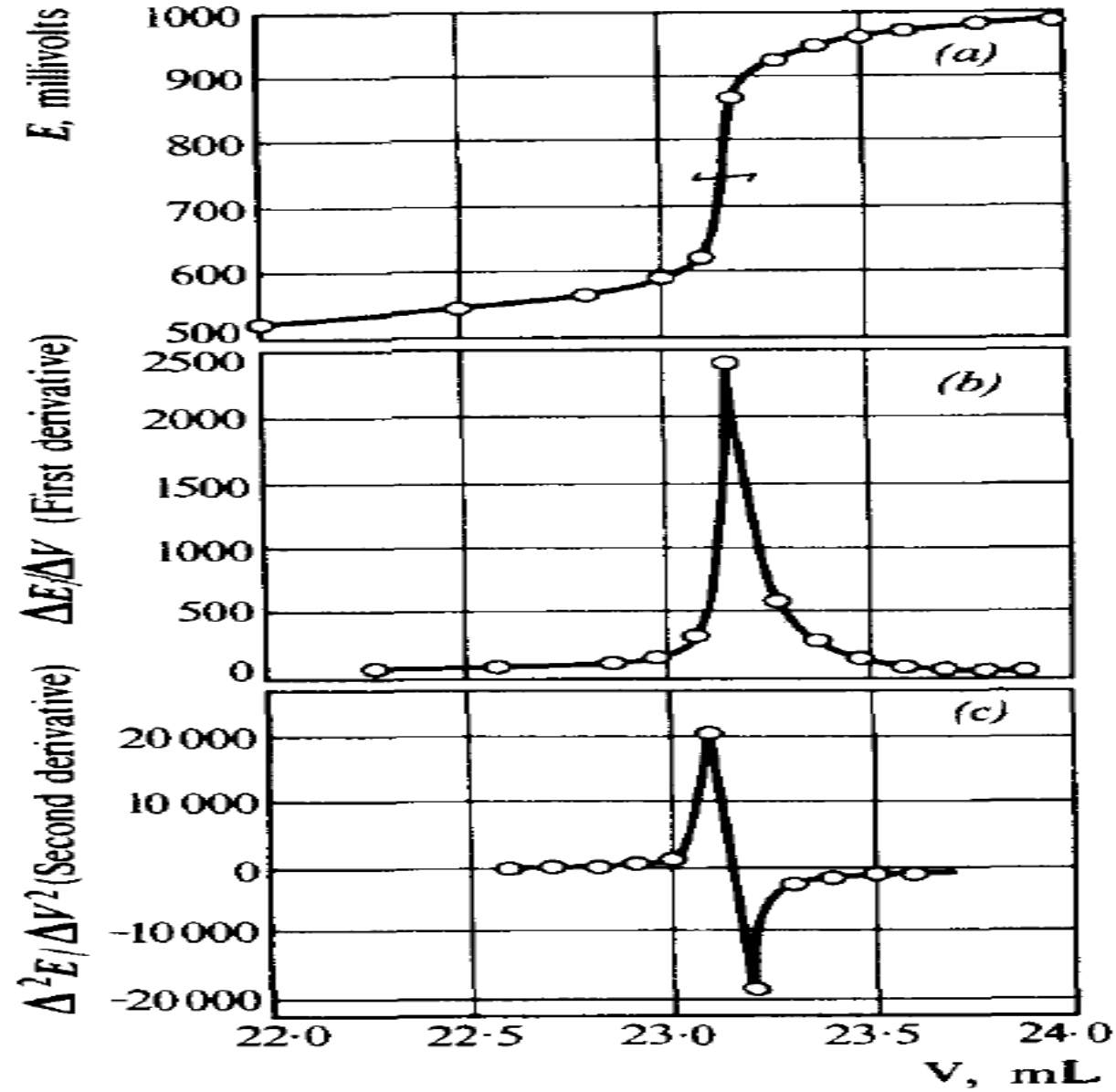
n - no. of electrons involved in the reaction

(ox) - conc. of oxidant

(Red) - conc. of reductant

- A titration curve has a characteristic sigmoid curve
- The part of the curve that has the **maximum change** marks the **equivalence point** of the titration
- In the case of an acid titration, the slope is increasing before the equivalence point and decreasing after the equivalence point (just the opposite is true for the titration of a base)
- Note that the slope changes **fastest just before and just after** the equivalence point

• At the equivalence point, it does not change at all, and we can



- Potentiometric titration of 25.0ml of 0.1 M ammonium iron(ii) sulphate with standard (0.1095M) cerium(iv) sulphate solution using platinum and saturated calomel electrodes:



- For locating the end point, only the experimental figures in the vicinity of the equivalence point are required.

Potentiometric titration of Fe^{2+} solution with 0.1095 M Ce^{4+} solution, using pt and calomel electrodes

Ce^{4+} solution added, V (mL)	E (mV)	$\Delta E/\Delta V$ (mV mL ⁻¹)	$\Delta^2 E/\Delta V^2$ (mV mL ⁻²)
22.00	523		
		40	
22.50	543		
		70	
22.60	550		0
		70	
22.70	557		100
		80	
22.80	565		200
		100	
22.90	575		500
		150	
23.00	590		1 500
		300	
23.10	620		21 000
		2400	
23.20	860		- 18 500
		550	
23.30	915		- 2 600
		290	
23.40	944		- 1 500
		140	
23.50	958		- 840
		56	
24.00	986		- 300

Where is the end point?

Quantitative Analysis using external standards, i.e calibration curve

Example

The concentration of Ca^{2+} in a water sample was determined by the method of external standards. The ionic strength of the samples and standards was maintained at a nearly constant level by making each solution 0.5 M in KNO_3 . The measured cell potentials for the external standards are shown in the following table.

$[\text{Ca}^{2+}]$ (M)	E_{cell} (V)
1.00×10^{-5}	-0.125
5.00×10^{-5}	-0.103
1.00×10^{-4}	-0.093
5.00×10^{-4}	-0.072
1.00×10^{-3}	-0.065
5.00×10^{-3}	-0.043
1.00×10^{-2}	-0.033

What is the concentration of Ca^{2+} in a water sample if its cell potential is found to be -0.084 V?

SOLUTION

Linear regression gives the equation for the calibration curve as

$$E_{\text{cell}} = 0.027 + 0.0303 \log [\text{Ca}^{2+}]$$

Substituting the cell potential for the sample gives the concentration of Ca^{2+} as 2.17×10^{-4} M. Note that the slope of the calibration curve is slightly different from the ideal value of $0.05916/2 = 0.02958$.

Measuring pH of solution

$$E_{\text{cell}} = K + 0.05916 \log [H^+]$$



Electrode

Thank you!!!